Lucas Gabardos, *PhD*

Experimental Physics Researcher

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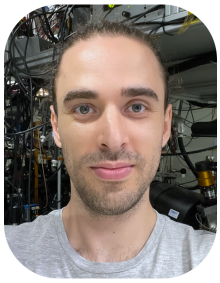

Experimental AMO physicist experienced with cold-atom setups, spin quantum systems, optimal control, lasers, optical lattices, vacuum technologies, experimental control and numerical simulations.

Work experience

Jul 2023 – Oct 2025 (2 years 4 months, full time)
Postdoctoral researcher, Nanyang Technological University, School of Physical and Mathematical Sciences, Centre for Quantum Technologies, Singapore
Team: Ultracold Gases and Nanophotonics (website)

- Transfer of a topological skyrmion from light to an atomic ensemble
  ↪ SLM laser beam shaping, atomic state coherent control
- Optimal control optimization of qutrit gates on hyperfine states of ^{87}Sr for a spin echo Ramsey sequence
  ↪ Optimal control theory, AWG $10\,\mu s$ pulse shaping

Dec 2020 – Nov 2022 (2 years, full time)
Postdoctoral researcher, Université Toulouse III - Paul Sabatier, Laboratoire Collisions Agrégats Réactivité (UMR CNRS 5589), Toulouse, France
Quantum engineering group (website), Cold atoms team

- Optimal control of a ^{87}Rb BEC in a one-dimensional optical lattice
- Nucleation of fractional momentum peaks in a Floquet system
- Transport of matter waves in a non-diffusive Hamiltonian ratchet

Education

Sep 2016 – Sep 2020
***Ph.D.* in Quantum Physics**, Université Sorbonne Paris Nord, Laboratoire de Physique des Lasers (UMR CNRS 7538), Villetaneuse, France
Team: Magnetic Quantum Gases (website)
Supervisors: Bruno Laburthe-Tolra (doctoral advisor) and Laurent Vernac
Thesis title: Out-of-equilibrium dynamics of spins coupled by dipolar interactions

- Cooling and compression of ^{52}Cr atoms using gray molasses
- Spin mixing and protection of ferromagnetism in a ^{52}Cr BEC
- Out-of-equilibrium quantum magnetism in a 3D lattice system
- Spin dynamics across a superfluid to Mott insulator transition

Sep 2014 – Jun 2016
***M.Sc.* in Quantum Devices**, Université Paris Diderot, Paris, France

Sep 2013 – Jun 2014
***B.Sc.* in Fundamental Physics**, Université Paris Diderot, Paris, France

Internships

Mar – Jun 2016 (4 months, full time)
Laboratoire de Physique des Lasers, Université Sorbonne Paris Nord, Villetaneuse, France
Team: Magnetic Quantum Gases, Supervisor: Laurent Vernac
Subject: Preparation of the internal state of ^{53}Cr atoms through magnetic dipolar transitions between hyperfine levels of their fundamental state – toward spin-resolved imaging

Jun – Jul 2015 **Laboratoire Kastler Brossel**, Université Pierre-et-Marie-Curie, Paris, France
2 months, full time
Team: Trapped Ions, Supervisors: Jean-Philippe Karr and Laurent Hilico
Subject: Study of a $626\,\mathrm{nm}$ DBR diode laser for cooling beryllium ions as part of a H_2^+ metrology experiment

Jun – Jul 2014 **Laboratoire Matériaux et Phénomènes Quantiques**, Université Paris Diderot, Paris, France
6 weeks, full time
Team: Trapped Ions and Quantum Information, Supervisor: Luca Guidoni
Subject: Study of the heating rate of an ion trapped in a planar Paul microtrap through the analysis of its fluorescence dynamics

Contributions to national and international conferences

Jul 2017 **Young Atom Opticians (YAO)**, Paris, France (poster)

Jul 2018 **International Conference on Atomic Physics (ICAP)**, Barcelona, Spain (poster)

Jul 2022 **Congrès de la société française d'optique**, yearly conference of the French optical society, Nice, France (poster)

Aug 2024 **Pacific Rim Conference on Lasers and Electro-Optics (CLEO-PR)**, Incheon, South Korea (talk)

Oct 2024 **Institute of Physics Singapore (IPS) Meeting**, Singapore (talk)

Experimental expertise

- Cold-atom experiments
- Optical lattices
- Laser cooling and trapping (2D and 3D MOT, dipole trap...)
- Ultra-high vacuum (UHV) systems
- Lasers and optical alignment (ECDL, TA, SHG, Fabry-Pérot cavities, AOM, EOM...)
- Control electronics (arbitrary wave generation, mixers, digital/analog signals...)

Computer skills

Scientific

Python	■■■■□	data analysis, numerical computation
Matlab	■■■■□	numerical computation, simulation
Mathematica	■■■■□	analytical and numerical computation, simulation
IGOR Pro	■■■□□	data acquisition and analysis during my PhD

Instrument control

Labview	■■□□□	used primarily for experiment control; limited programming

Pictures and words

LaTeX	■■■■□	thesis and article production
Microsoft Office	■■■■□	Word, Powerpoint and Excel
Inkscape	■■■□□	basic vector drawing

Computer-assisted design

FreeCAD	■■■□□	3D design

Languages

French	■■■■■	Native language	
English	■■■■■	Professional working proficiency	*TOEIC listening and reading 495/495 (2019)*
Chinese	□□□□□	Very early beginner	

List of publications

2018 – PRA **_Spin mixing and protection of ferromagnetism in a spinor dipolar condensate_**
S. Lepoutre, K. Kechadi, B. Naylor, B. Zhu, L. Gabardos, L. Isaev, P. Pedri, A. M. Rey, L. Vernac, and B. Laburthe-Tolra
Abstract: We study spin mixing dynamics in a chromium dipolar Bose-Einstein condensate, after tilting the atomic spins by an angle θ with respect to the magnetic field. Spin mixing is triggered by dipolar coupling, but, once dynamics has started, it is mostly driven by contact interactions. For the particular case $\theta = \pi/2$, an external spin-orbit coupling term induced by a magnetic gradient is required to enable the dynamics. Then the initial ferromagnetic character of the gas is locally preserved, an unexpected feature that we attribute to large spin-dependent contact interactions.
Physical Review A 97, 023610 (2018)

2018 – PRL **_Collective spin modes of a trapped quantum ferrofluid_**
S. Lepoutre, L. Gabardos, K. Kechadi, P. Pedri, O. Gorceix, E. Maréchal, L. Vernac, and B. Laburthe-Tolra
Abstract: We report on the observation of a collective spin mode in a spinor Bose-Einstein condensate. Initially, all spins point perpendicular to the external magnetic field. The lowest energy mode consists of a sinusoidal oscillation of the local spin around its original axis, with an oscillation amplitude that linearly depends on the spatial coordinates. The frequency of the oscillation is set by the zero-point kinetic energy of the BEC. The observations are in excellent agreement with hydrodynamic equations. The observed spin mode has a universal character, independent of the atomic spin and spin-dependent contact interactions.
Physical Review Letters 121, 013201 (2018)

2019 – PRA **_Cooling all external degrees of freedom of optically trapped chromium atoms using gray molasses_**
L. Gabardos, S. Lepoutre, O. Gorceix, L. Vernac, and B. Laburthe-Tolra
Abstract: We report on a scheme to cool and compress trapped clouds of highly magnetic ^{52}Cr atoms. This scheme combines sequences of gray molasses, which freeze the velocity distribution, and free evolutions in the (close to) harmonic trap, which periodically exchange the spatial and velocity degrees of freedom. Taken together, the successive gray molasses pulses cool all external degrees of freedom, which leads to an increase of the phase-space density (PSD) by a factor of ≈ 250, allowing to reach a high final PSD of $\approx 1.7 \times 10^{-3}$. These experiments are performed within an optical dipole trap, in which the gray molasses works as well as it does in free space. The obtained samples are then an ideal starting point for the evaporation stage aiming at the quantum regime.
Physical Review A 99, 023607 (2019)

2019 – Nature Com. **_Out-of-equilibrium quantum magnetism and thermalization in a spin-3 many-body dipolar lattice system_**
S. Lepoutre, J. Schachenmayer, L. Gabardos, B. Zhu, B. Naylor, E. Maréchal, O. Gorceix, A. M. Rey, L. Vernac and B. Laburthe-Tolra
Abstract: Understanding quantum thermalization through entanglement build up in isolated quantum systems addresses fundamental questions on how unitary dynamics connects to statistical physics. Spin systems made of long-range interacting atoms offer an ideal experimental platform to investigate this question. Here, we study the spin dynamics and approach towards local thermal equilibrium of a macroscopic ensemble of $S = 3$ chromium atoms pinned in a three-dimensional optical lattice and prepared in a pure coherent spin state, under the effect of magnetic dipole–dipole interactions. Our isolated system thermalizes under its own dynamics, reaching a steady state consistent with a thermal ensemble with a temperature dictated from the system's energy. The build up of quantum correlations during the dynamics is supported by comparison with an improved numerical quantum phase-space method. Our observations are consistent with a scenario of quantum thermalization linked to the growth of entanglement entropy.
Nature Communications 10, 1714 (2019)

2019 – PRA **_Dynamics of an itinerant spin-3 atomic dipolar gas in an optical lattice_**

P. Fersterer, A. Safavi-Naini, B. Zhu, L. Gabardos, S. Lepoutre, L. Vernac, B. Laburthe-Tolra, P. Blair Blakie, and A. M. Rey

Abstract (shortened): [...] In this work we report on our investigation of the quantum many-body dynamics of a large ensemble of bosonic magnetic chromium atoms with spin $S = 3$ in a three-dimensional lattice as a function of lattice depth. Using extensive theory and experimental comparisons, we study the dynamics of the population of the different Zeeman levels and the total magnetization of the gas across the superfluid to the Mott insulator transition. We are able to identify two distinct regimes. At low lattice depths, where atoms are in the superfluid regime, we observe that the spin dynamics is strongly determined by the competition between particle motion, on-site interactions, and external magnetic-field gradients. Contact spin-dependent interactions help to stabilize the collective spin length, which sets the total magnetization of the gas. On the contrary, at high lattice depths, transport is largely frozen out. In this regime, while the spin populations are mainly driven by long-range dipolar interactions, magnetic-field gradients also play a major role in the total spin demagnetization. [...]

Physical Review A 100, 033609 (2019)

2020 – PRL **_Relaxation of the collective magnetization of a dense 3D array of interacting dipolar S = 3 atoms_**

L. Gabardos, B. Zhu, S. Lepoutre, A. M. Rey, B. Laburthe-Tolra, and L. Vernac

Abstract: We report on measurements of the dynamics of the total magnetization and spin populations in an almost unit-filled lattice system comprising about 10^4 spin $S = 3$ chromium atoms, under the effect of dipolar interactions. The observed spin population dynamics is unaffected by the use of a spin echo and fully consistent with numerical simulations of the $S = 3$ XXZ spin model. On the contrary, the observed magnetization decays slower than in simulations and, surprisingly, reaches a small but nonzero asymptotic value within the longest timescale. Our findings show that spin coherences are sensitive probes to systematic effects affecting quantum many-body behavior that cannot be diagnosed by merely measuring spin populations.

Physical Review Letters 125, 143401 (2020)

2021 – PRX Quantum **_Quantum state control of a Bose-Einstein condensate in an optical lattice_**

N. Dupont, G. Chatelain, L. Gabardos, M. Arnal, J. Billy, B. Peaudecerf, D. Sugny, and D. Guéry-Odelin

Abstract: We report on the efficient design of quantum optimal-control protocols to manipulate the motional states of an atomic Bose-Einstein condensate (BEC) in a one-dimensional optical lattice. Our protocols operate on the momentum comb associated with the lattice. In contrast to previous works also dealing with control in discrete and large Hilbert spaces, our control schemes allow us to reach a wide variety of targets by varying a single parameter, the lattice position. With this technique, we experimentally demonstrate a precise, robust, and versatile control: we optimize the transfer of the BEC to a single or multiple quantized momentum states with full control on the relative phase between the different momentum components. This also allows us to prepare the BEC in a given eigenstate of the lattice band structure, or superposition thereof.

Physical Review X Quantum 2, 040303 (2021)

2023 – NJP **_Phase-space distributions of Bose-Einstein condensates in an optical lattice: optimal shaping and reconstruction_**

N. Dupont, F. Arrouas, L. Gabardos, N. Ombredane, J. Billy, B. Peaudecerf, D. Sugny and D. Guéry-Odelin

Abstract: We apply quantum optimal control to shape the phase-space distribution of Bose-Einstein condensates in a one-dimensional optical lattice. By a time-dependent modulation of the lattice position, determined from optimal control theory, we prepare, in the phase space of each lattice site, translated and squeezed Gaussian states, and superpositions of Gaussian states. Complete reconstruction of these non-trivial states is performed through a maximum likelihood state tomography. As a practical application of our method to quantum simulations, we initialize the atomic wavefunction in an optimal Floquet-state superposition to enhance dynamical tunneling signals.

New Journal of Physics 25, 013012 (2023)

2023 – PNAS **_Emergence of tunable periodic density correlations in a Floquet–Bloch system_**
N. Dupont, L. Gabardos, F. Arrouas, G. Chatelain, M. Arnal, J. Billy, P. Schlagheck, B. Peaudecerf and D. Guéry-Odelin
Abstract: In quantum gases, two-body interactions are responsible for a variety of instabilities that depend on the characteristics of both trapping and interactions. These instabilities can lead to the appearance of new structures or patterns. We report on the Floquet engineering of such a parametric instability, on a Bose–Einstein condensate held in a time-modulated optical lattice. The modulation triggers a destabilization of the condensate into a state exhibiting a density modulation with a new spatial periodicity. This new crystal-like order, which shares characteristic correlation properties with a supersolid, directly depends on the modulation parameters: The interplay between the Floquet spectrum and interactions generates narrow and adjustable instability regions, leading to the growth, from quantum or thermal fluctuations, of modes with a density modulation noncommensurate with the lattice spacing. This study demonstrates the production of metastable exotic states of matter through Floquet engineering and paves the way for further studies of dissipation in the resulting phase and of similar phenomena in other geometries.
Proceedings of the National Academy of Sciences of the U.S.A. 120 (32), e2300980120 (2023)

2023 – PRL **_Hamiltonian Ratchet for Matter-Wave Transport_**
N. Dupont, L. Gabardos, F. Arrouas, N. Ombredane, J. Billy, B. Peaudecerf and D. Guéry-Odelin
Abstract: We report on the design of a Hamiltonian ratchet exploiting periodically at rest integrable trajectories in the phase space of a modulated periodic potential, leading to the linear nondiffusive transport of particles. Using Bose-Einstein condensates in a modulated one-dimensional optical lattice, we make the first observations of this spatial ratchet, which provides way to coherently transport matter waves with possible applications in quantum technologies. In the semiclassical regime, the quantum transport strongly depends on the effective Planck constant due to Floquet state mixing. We also demonstrate the interest of quantum optimal control for efficient initial state preparation into the transporting Floquet states to enhance the transport periodicity.
Physical Review Letters 131, 133401 (2023)

2024 – Comptes Rendus. Physique **_Floquet operator engineering for quantum state stroboscopic stabilization_**
F. Arrouas, N. Ombredane, L. Gabardos, E. Dionis, N. Dupont, J. Billy, B. Peaudecerf, D. Sugny and D. Guéry-Odelin
Abstract: Optimal control is a valuable tool for quantum simulation, allowing for the optimized preparation, manipulation, and measurement of quantum states. Through the optimization of a time-dependent control parameter, target states can be prepared to initialize or engineer specific quantum dynamics. In this work, we focus on the tailoring of a unitary evolution leading to the stroboscopic stabilization of quantum states of a Bose–Einstein condensate in an optical lattice. We show how, for states with space and time symmetries, such an evolution can be derived from the initial state-preparation controls; while for a general target state we make use of quantum optimal control to directly generate a stabilizing Floquet operator. Numerical optimizations highlight the existence of a quantum speed limit for this stabilization process, and our experimental results demonstrate the efficient stabilization of a broad range of quantum states in the lattice.
Comptes Rendus. Physique, CNRS Gold Medal Jean Dalibard, Volume 24 (2023), pp. 173-185

2025 – APL Photonics **_Topological optical skyrmion transfer to matter_**
C. Mitra, C. S. Madasu, L. Gabardos, C. C. Kwong, Y. Shen, J. Ruostekoski and D. Wilkowski
Abstract: The ability of structured light to mimic exotic topological skyrmion textures encountered in high-energy physics, cosmology, magnetic materials, and superfluids has recently received considerable attention. Despite their promise as mechanisms for data encoding and storage, there has been a lack of studies addressing the transfer and storage of the topology of optical skyrmions to matter. Here, we demonstrate a high-fidelity mapping of skyrmion topology from a laser beam onto a gas of cold atoms, where it is detected in its new non-propagating form. Within the spatial overlap of the beam and atom cloud, the skyrmion topological charge is preserved, with a reduction from $Q \approx 0.91$ to $Q \approx 0.84$, mainly due to the beam width exceeding the sample size. Our work potentially opens novel avenues for topological photonic state storage and the analysis of more complex structured light topologies.
Applied Physics Letters Photonics 10, 046113 (2025)

2025 – Nature Com. ***Experimental realization of a SU(3) color-orbit coupling in an ultracold gas***
C. S. Madasu, C. Mitra, L. Gabardos, K. D. Rathod, T. Zanon-Willette, C. Miniatura, F. Chevy, C. C. Kwong and D. Wilkowski
Abstract: Spin-orbit interaction couples the spin of a particle to its motion and leads to spin-induced transport phenomena such as spin-Hall effects and Chern insulators. In this work, we extend the concept of internal-external state coupling to higher internal symmetry, exploring features beyond the established spin-orbit regime. We couple suitable resonant laser beams to a gas of ultracold atoms, thereby inducing artificial SU(3) non-Abelian gauge fields that act on a degenerate ground state manifold comprised of three dark states. We demonstrate the inherent all-state connectivity of SU(3) systems by performing targeted geometric transformations. Then, we investigate color-orbit coupling, an extension of SU(2) spin-orbit coupling to SU(3) systems. We reveal a rich dynamical interplay between three distinct oscillation frequencies, which possesses interesting analogies with neutrino oscillations and quark mixing mechanisms. In the future, the system should provide a testbed for further investigation of topological properties of SU(3) systems.
Nature Communications 16, 8448 (2025)

2026 One paper planned on the optimal control optimization of qutrit gates on hyperfine states of ^{87}Sr